

Topic C2

Atoms, elements and compounds

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

Syllabus map

c2.1	Elements, compounds and mixtures
c2.2	Atomic structure and the Periodic Table
c2.3	Isotopes
c2.4	Ions and ionic bonds
c2.5	Simple molecules and covalent bonds
c2.6	Giant covalent structures
c2.7	Metallic bonding

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

1. Elements, compounds and mixtures

An element contains one type of atom. A compound contains two or more elements chemically combined in fixed proportions and can be separated only chemically. A mixture contains substances not chemically combined; its components retain their properties and can be separated physically.

Examiner insight

Use “atoms of one element” rather than “one atom” when defining an element.

2. Atomic structure

Protons and neutrons occupy the nucleus; electrons occupy shells. Proton number identifies the element. Nucleon number is protons plus neutrons. In a neutral atom, electrons equal protons. The first shell holds 2 electrons and subsequent syllabus-level arrangements are written, for example, 2,8,7.

neutrons = nucleon number - proton number

Worked example

For chlorine-37: protons 17, electrons 17, neutrons $37 - 17 = 20$.

Examiner insight

Chemical behaviour depends mainly on outer-shell electrons.

3. Isotopes and relative atomic mass

Isotopes are atoms of the same element with the same proton number but different neutron numbers. They have nearly identical chemical properties because they have the same electronic structure. Relative atomic mass is the weighted mean mass of naturally occurring isotopes relative to 1/12 of carbon-12.

$A_r = \text{sum}(\text{isotopic mass} \times \text{abundance}) / 100$

Worked example

75% chlorine-35 and 25% chlorine-37 gives $A_r = (35 \times 75 + 37 \times 25) / 100 = 35.5$.

Examiner insight

Use abundances as weights; do not simply average isotope masses.

4. Ionic bonding

Metals form positive ions by losing electrons; non-metals form negative ions by gaining electrons. Ionic bonding is strong electrostatic attraction between oppositely charged ions in a giant lattice. Ionic compounds have high melting points and conduct only when molten or aqueous because ions must be mobile.

Worked example

Magnesium loses two electrons to form Mg^{2+} ; each chlorine gains one, so the formula is MgCl_2 .

Examiner insight

“Electron transfer” explains ion formation; “electrostatic attraction” explains the bond.

5. Covalent structures

A covalent bond is a shared pair of electrons. Simple molecular substances have strong covalent bonds within molecules but weak intermolecular forces, so they often have low melting and boiling points and do not conduct. Giant covalent structures have many strong bonds and very high melting points.

Examiner insight

Graphite conducts because one electron per carbon is delocalised; diamond does not and is hard because every carbon forms four bonds.

6. Metallic bonding

Metals are giant lattices of positive ions in a sea of delocalised electrons. Strong electrostatic attraction holds the structure together. Mobile electrons conduct electricity and thermal energy; layers of ions can slide, explaining malleability and ductility.

Examiner insight

Do not describe metallic bonding as positive ions attracted to negative ions; the electrons are delocalised.

Exam technique

High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

Practice Paper

Topic C2

toms, elements and compounds

Instructions

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement defines a compound? [1]
 - A One type of atom only
 - B Elements chemically combined in fixed proportions
 - C Substances mixed in any proportion
 - D A substance separable by filtration

2. An atom has proton number 12 and nucleon number 25. How many neutrons? [1]
 - A 12
 - B 13
 - C 25
 - D 37

3. Which particles determine an element's chemical properties? [1]
 - A nuclear neutrons
 - B outer-shell electrons
 - C all nucleons
 - D only inner electrons

4. Isotopes of one element have different numbers of [1]
 - A protons
 - B outer shells
 - C neutrons
 - D nuclear charges

5. What is the formula of aluminium oxide from Al^{3+} and O^{2-} ? [1]
 - A AlO
 - B Al_2O_3
 - C Al_3O_2
 - D AlO_3

6. What is an ionic bond? [1]
 - A a shared electron pair
 - B attraction between nuclei
 - C electrostatic attraction between opposite ions
 - D attraction between molecules

7. Why does molten sodium chloride conduct? [1]
 - A Electrons leave atoms.
 - B Ions are mobile.
 - C Molecules are volatile.
 - D Protons move.

8. Which substance has a giant covalent structure? [1]
- A methane
 - B iodine
 - C diamond
 - D water
9. Why does graphite conduct electricity? [1]
- A It contains mobile ions.
 - B It has delocalised electrons.
 - C Its molecules are weakly bonded.
 - D Carbon atoms are charged.
10. Which model describes a metal? [1]
- A molecules with weak forces
 - B positive ions and delocalised electrons
 - C negative ions in a lattice
 - D isolated neutral atoms

Section B · Structured questions

11 An atom of potassium has proton number 19 and nucleon number 39.

(a) State the numbers of protons, neutrons and electrons. [3]

(b) Give its electronic structure. [1]

(c) Predict the ion formed and explain how it forms. [2]

(d) Explain why potassium and sodium have similar reactions. [2]

[Total: 8]

12 Chlorine contains 75% chlorine-35 and 25% chlorine-37.

(a) Define isotope. [2]

(b) Calculate the relative atomic mass. [3]

(c) Explain why the isotopes react similarly. [2]

(d) State one use of an isotope. [1]

[Total: 8]

13 Magnesium reacts with chlorine to form magnesium chloride.

(a) Describe electron transfer during ion formation. [3]

(b) Deduce the formula of the compound. [1]

(c) Explain its high melting point. [2]

(d) Explain why the solid does not conduct but the melt does. [2]

[Total: 8]

14 Diamond and graphite are allotropes of carbon.

(a) Describe the bonding in diamond. [2]

(b) Explain why diamond is hard. [2]

(c) Explain why graphite is soft. [2]

(d) Explain why graphite conducts.

[2]

[Total: 8]

15 Compare a simple molecular substance with a metal.

(a) Describe bonding in a simple molecule.

[2]

(b) Explain its low boiling point.

[2]

(c) Describe metallic bonding.

[2]

(d) Explain metal conductivity and malleability.

[2]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

Solutions · Multiple choice

- 1 B. A compound contains chemically bonded elements in fixed ratios.
- 2 B. $25 - 12 = 13$.
- 3 B. Bonding and reactions involve valence electrons.
- 4 C. They share proton number but differ in neutrons.
- 5 B. Charges balance at +6 and -6.
- 6 C. This attraction holds the ionic lattice.
- 7 B. Charge is carried by mobile ions.
- 8 C. Diamond is a continuous carbon network.
- 9 B. Delocalised electrons move through layers.
- 10 B. That is metallic bonding.

Solutions · Structured questions

11. 11(a) 19 protons, 20 neutrons, 19 electrons [3]. (b) 2,8,8,1 [1]. (c) K^+ by loss of one electron [2]. (d) Both have one outer-shell electron [1], so form +1 ions similarly [1].

12. 12(a) Same proton number but different neutron number [2]. (b) $(35 \times 75 + 37 \times 25) / 100 = 35.5$ [3]. (c) Same electron arrangement / same outer electrons [2]. (d) Any valid tracer, medical or dating use [1].

13. 13(a) Mg loses two electrons; two Cl atoms each gain one; Mg^{2+} and Cl^- form [3]. (b) $MgCl_2$ [1]. (c) Strong electrostatic attraction throughout giant lattice requires much energy [2]. (d) Ions fixed in solid but mobile in melt [2].

14. 14(a) Each carbon covalently bonded to four in a giant structure [2]. (b) Many strong covalent bonds in all directions [2]. (c) Weak forces between layers allow sliding [2]. (d) One delocalised electron per carbon carries charge [2].

15. 15(a) Atoms share electron pairs [2]. (b) Only weak intermolecular forces are overcome [2]. (c) Attraction between positive ions and delocalised electrons [2]. (d) Mobile electrons conduct; layers of ions slide without breaking bonding [2].

Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.